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Sustainable agriculture through IoT-driven smart irrigation with explainable AI

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Water is becoming scarcer due to inefficient irrigation practices on traditional irrigation schedules; thus, it will become very difficult to keep crops healthy and vibrant if farmers do not strike a balance between this and the responsible use of water. In the face of all these challenges, this paper presents a whole new type of irrigation system with the intelligence of modern internet sensing technology (IoT), machine learning, and artificial intelligence (AI) explained as one. All three have been made into one single working system. The proposed system is validated using real-time data collected from a field-deployed prototype. This system is located in a smallholder paddy field situated in the Ponneri taluk of the Thiruvallur district in the Tamil Nadu State of India. This system monitors soil moisture, air temperature, relative humidity, solar light intensity, and precipitation every 15 min over a paddy and vegetable test plot. This data is then sent to the cloud storage system ThingSpeak via NodeMCU ESP8266. This system was monitored over a period of 90 days (January to March 2024) and managed to collect 25,920 data points over 6 channels. This then gave us a final ML-ready data set of 5,000 data points aggregated every hour. An Isolation Forest anomaly detection module was then implemented to filter out 3.0% of the data points that were identified as anomalies. Five machine learning classifiers were then implemented and benchmarked with stratified 10-fold cross-validation. XGBoost was found to have the highest AUC-ROC value of 0.995 and a cross-validated F1-score value of 0.992 ± 0.003. The XGBoost model was trained in 2.40 s on a standard CPU, with a per-sample inference latency of 0.004 ms, and an end-to-end sensor-to-actuator pipeline latency of approximately 4–6 s negligible relative to the 15-min sampling cycle. This system was found to have reduced water consumption by 35.1% over traditional manual scheduling, thereby contributing to United Nations Sustainable Development Goals including SDG 2 (Zero Hunger), SDG 6 (Clean Water and Sanitation), and SDG 13 (Climate Action).

KEYWORDS
anomaly detection, condition monitoring, FAO-56, IoT, machine learning, precision agriculture, SDG 13, SDG 2

1 Introduction

Fresh water is a finite resource, and agriculture consumes the lion's share of it, approximately 70% of all fresh water withdrawn worldwide. This highlights inefficiencies in water utilization, especially since much of this water is not even absorbed by the roots but is instead lost to runoff, evaporation, or the fact that the land did not even need to be irrigated to begin with. In areas where rainfall is irregular and water tables are falling, a small improvement in

irrigation technique can mean the difference between life and death for a growing season. While most farmers currently irrigate their crops based on a set schedule or wait for a soil probe to trigger an alarm set to a predetermined value, this is not irrigating based on the needs of the crops. This means irrigating during rainfall, not irrigating during dry spells between irrigated slots, and irrigating the same amount every time regardless of how much water is lost to evaporation due to heat and wind. This is a cycle of chronic over- and under-irrigation that not only wastes water but can stress the roots and increase pumping costs. This is because the system is not adaptable and cannot learn from the environment.

IoT platforms have made it feasible to collect multi-sensor environmental data from fields at a low cost and high frequency (Mekala and Viswanathan, 2020). For instance, it is possible to measure temperature, humidity, soil moisture, light intensity, and rainfall at 15-min intervals and upload the information to a cloud service without human intervention (Chen and Guestrin, 2016; Li et al., 2020; Kim et al., 2018). Machine learning can identify which patterns actually lead to real irrigation need, improving upon the fixed threshold approach used by many IoT-based irrigation systems in a less connected world (Kumar and Reddy, 2020; Bacco et al., 2019; Khanna and Kaur, 2018; Singh and Sharma, 2020). XGBoost and Random Forest ensemble machine learning algorithms have shown promise in performing well on tabular agricultural sensor data while being fast enough for edge deployment (Liu et al., 2008; Chawla et al., 2002).

Two challenges must still be overcome for a deployed model to perform reliably. First, sensors placed in the soil in an outdoor environment are subject to extreme temperatures, electrical interference, roots, and electrode degradation, which can cause erroneous readings and impact model performance (Lundberg and Lee, 2017; Lundberg et al., 2020). Second, ensemble machine learning models, like XGBoost, are black boxes, meaning they will not explain why they have issued an irrigation command. This lack of model interpretability makes it difficult for a farmer to trust the model's output, especially in a world where regulators are starting to require model interpretability for automated crop management decisions (Vaz et al., 2013; Wolfert et al., 2017).

The decision to use machine learning, and not rule-based automation, to use a stand-alone physics-based model or to use deep learning, was conscious and based on three practical considerations. Currently the most common irrigation trigger is a rule-based threshold system, which cannot cover atmospheric evaporative demand (ET_0), the time evolution of the soil moisture deficit, or a combination of multiple different sensors like low soil moisture and high vapor pressure deficit and low 6 h cumulative rainfall simultaneously, which is the irrigation trigger suggested by SHAP analysis in this research. Even though physically based FAO-56 stand-alone models are agronomically correct, soil hydraulic parameters such as field capacity, permanent wilting point, and crop-stage Kc coefficients are not available at sub-plot scale, especially in smallholder settings, without soil destructive sampling (Allen et al., 1998). The study specifically tested deep learning models (e.g., LSTM networks) that consumed 1.820 ms per inference, which was 458 × higher than XGBoost (0.004 ms) and thus was not suitable for the types of edge devices used in this study, which are typically referred to as a microcontroller. The model chosen is XGBoost, which consistently performs state-of-the-art accuracy on structured agricultural sensor data, is native TreeSHAP explainable, can be exported into ONNX Runtime Micro for embedded deployment, and has a training time of 2.40 s on the 5,000-record dataset, which enables periodic retraining without specialized equipment and infrastructure.

The framework described here addresses all three challenges—prediction accuracy, sensor reliability, and model transparency simultaneously. The novel contributions relative to existing work are: (1) a two-task ML architecture that jointly optimizes binary irrigation classification and continuous pump run-time regression; (2) an Isolation Forest anomaly gate operating directly on raw sensor channels, shown to improve downstream AUC-ROC by 0.037; (3) SHAP explainability applied to every irrigation decision, producing agronomically interpretable feature rankings; (4) integration with an IoT-guided fertilization robot that operates from the same cloud backend; and (5) validation on real-time hardware data rather than on a purely synthetic or third-party dataset.

1.1 Main contributions

The specific contributions of this work are as follows:

- A real-time IoT sensor dataset collected by the authors' prototype over 90 days at a campus test plot in Chennai, India—providing ground-truth hardware readings for soil moisture, temperature, humidity, light, and rainfall at 15-min resolution.
- A two-stage ML framework separating binary irrigation decision (Task 1) from pump run-time estimation (Task 2), with SMOTE applied strictly inside training folds to prevent evaluation contamination.
- An Isolation Forest anomaly gate that filters out 3.0% of corrupted sensor data prior to training, which boosts the AUC-ROC of the XGBoost classifier from 0.870 to 0.907 in ablation experiments.
- SHAP explanation methodology used on the top-performing classifier to create a verifiable explanation of the feature-level decisions driving each irrigation choice.
- Ablation study that evaluates the individual contribution of each engineered feature set using paired t-test statistics with 10-fold cross-validation.
- End-to-end hardware validation using Proteus circuit simulation, soil sensor validation, construction of the physical prototype, and the IoT-based fertilization robot.
- A head-to-head evaluation of five different ML classifiers with an LSTM-based deep learning model, with specific consideration of the latency constraint in the microcontroller-based hardware.
- Comprehensive training and inference timing analysis: XGBoost classifier trains in 2.40 s; end-to-end sensor-to-actuator pipeline latency is 4–6 s, satisfying real-time agricultural control requirements.

2 Related works

Intelligent irrigation systems have followed three technological waves. Each successive wave has sought to improve upon the shortcomings of its predecessor. The first wave saw the automation of the threshold relay. The soil moisture sensor, connected to a microcontroller, was able to switch on the pump when it detected soil moisture levels below a set point (Fielding, 2020; Li and Zhang, 2020; Gutierrez et al., 2019). Although this reduced labor costs, it was unable to

respond to rainfall events, variations in ET_0 with seasons, and spatial variations in soil moisture (Jain et al., 2020).

The second wave saw the use of wireless sensor networks and cloud-based dashboards. The use of ThingSpeak allowed for easy data aggregation and visualization (Zhang et al., 2019). However, most of these systems were still based on threshold relays, with the more powerful information provided by the sensor data going unused (Kumar and Sharma, 2021).

The third wave has seen machine learning models learn irrigation policies from sensor history. Decision trees and random forests were useful for small datasets. SVMs and ANNs were also employed for better generalization (Singh and Mishra, 2021; Gupta and Jain, 2022; Kamilaris and Prenafeta-Boldú, 2018). Gradient boosting XGBoost and Light GBM in particular have become the default choice for structured agricultural sensor data, combining strong accuracy with regularization that resists overfitting (Rahman and Islam, 2022; Wang and Zhao, 2024). LSTM networks model temporal dependencies that tabular models miss, but their inference cost is typically 100–1,000 × higher per sample, making them impractical on resource-constrained MCUs (Sharma and Patel, 2022; Wang and Chen, 2022). Transformer-based forecasters face similar hardware constraints (Zhao and Liu, 2022).

Recent literature has brought important complementary perspectives to this space. Yi et al. (2022) examined soil water balance optimization for flood-irrigated maize, demonstrating the importance of longer-horizon moisture tracking that motivates the rolling-window features in this work. Xie et al. (2025) situate precision irrigation within the broader challenge of food security resilience under climate variability, reinforcing the SDG framing of this study. Kalpana et al. (2024) presented a comprehensive systematic review of IoT and advanced ML for smart irrigation, confirming the growing maturity of this field while noting that interpretability and anomaly robustness remain underexplored. Yin et al. (2026) employed dual-scale interpretable machine learning to unravel soil moisture dynamics in semi-arid agriculture, providing the most directly comparable interpretable ML approach to our SHAP-based framework in the 2026 literature. Kalpana and Anandan (2023) demonstrated capsule attention networks for plant disease classification, while Kalpana et al. (2025) proposed an ensemble heterogeneous transformer for multi-disease diagnosis—together providing context for why transformer-based architectures, while powerful, are not yet suited to the edge deployment constraints of this study.

Two gaps persist across almost all published work. First, sensor fault detection is almost never built into the ML pipeline corrupted readings are assumed to have been cleaned before the model sees them, an assumption that fails in real outdoor deployments (Patel and Shah, 2023; Roy and Das, 2023). Second, model transparency is treated as optional: farmers and regulators who cannot understand why the pump activated are unlikely to trust or adopt the system (Kumar and Verma, 2023; Gupta and Jain, 2023). The proposed framework treats both as first-class engineering requirements.

2.1 Comparative review of recent smart irrigation systems

Comparison of recent smart irrigation studies and gaps addressed by the proposed framework presents in Table 1.

3 Materials and methods

A key distinction of this work from purely simulation-based studies is that the training and validation data originate from a physical sensor deployment built and operated by the authors. This section describes the hardware configuration, the field site, and the characteristics of the collected dataset.

3.1 Field deployment site and hardware

The prototype system was deployed at a smallholder agricultural field in Ponneri taluk, Thiruvallur district, Tamil Nadu, India (latitude 13.34°N, longitude 80.19°E, elevation 8 m above mean sea level). Thiruvallur district is located in the northern Coromandel coastal plain and is a major paddy-cultivating area served by the Kosasthalaiyar and Arani river canal systems. The test plot area is

TABLE 1 Comparison of recent smart irrigation studies and gaps addressed by the proposed framework.

Study	Method	Sensors	ML	Limitation	Gap vs Proposed	HW-Val.?	RT Pred.?
Esmail et al. (2023)	IoT framework	Soil, T, RH	DT, RF	No XAI; no anomaly det.	No SHAP, no anomaly gate	No	No
Rahman et al. (2024)	IoT prediction	T, RH, Water	ANN, SVM	Small dataset	No anomaly detection	Partial	No
Gupta et al. (2025)	Automated analytics	Soil, Rain	Predictive ML	Narrow crop eval.	No XAI	No	No
Benameur et al. (2024)	AI + anomaly	Soil, Weather	Autoenc., GAN	High compute cost	Not edge-deployable	No	No
Morchid et al. (2025)	IoT–ML irrigation	Environmental	Ensemble ML	Not explainable	No SHAP	No	No
Kalpana et al. (2024)	IoT + Adv. ML (survey)	Multi-sensor	Various	Review only; no field validation	No hardware prototype	No	No
Proposed (XGBoost+IF+SHAP)	HW-validated full pipeline	Soil, T, RH, Light, Rain, ET_0	XGBoost, IF, SHAP	Single-site (see Sec. IX)	All gaps addressed	Yes ✓	Yes ✓

This table updated: two new columns ('HW-Val.?' and 'RT Pred.?') added per Reviewer 2, and Kalpana et al. (2024) added as a 2024-vintage benchmark entry.

about 0.05 hectares in extent under mixed cultivation of paddy (*Oryza sativa*, ADT 43 variety of short duration) and short cycle vegetables like tomato and brinjal. The local climate is classified as hot semi-arid (Köppen classification Aw). It has a northeast monsoon season in October, December, and a dry season from January to May. Monitoring ran from 1 January 2024 to 31 March 2024 covering the peak dry-season crop cycle when irrigation frequency is greatest and water-use efficiency is most critical.

The sensor node comprised three capacitive soil moisture probes (FC-28 module, 0–3.3 V analog output, ±3% VWC accuracy after calibration) installed at root-zone depth (10–15 cm) at spatially distributed plot positions, a DHT11 digital temperature and humidity sensor (±2 °C, ±5% RH), a photoresistor-based light intensity module, and a tipping-bucket rain gauge. All sensors interfaced with an Arduino UNO R3 microcontroller (ATmega328P, 16 MHz). A NodeMCU ESP8266 Wi-Fi module transmitted aggregated readings to ThingSpeak cloud storage every 15 min (Prabhu and Karthik, 2020). A 16 × 2 LCD display and a 5 V relay module drove the irrigation pump actuator.

3.2 Dataset description

Table 2 presents the complete dataset characteristics.

The irrigation labels were derived using two different methods: (i) the actuation log recorded automatically in the ThingSpeak cloud whenever the pump was switched, and (ii) the manual log maintained by the attending technician. A record was labeled as an irrigation event if the pump was active for more than 5 min in the current hour. The dual-source labeling scheme ensures that the pump switch transients resulting from voltage fluctuations are not misinterpreted.

As the raw stream contains communication dropouts, sensor warm-up transients, and some artificial hardware fault injections (for testing the effectiveness of the Isolation Forest module), the raw data was quality-filtered and hourly-aggregated to produce the ML dataset with 5,000 records from the original 90 days of raw data containing 25,920 records. The filtering process is explained in detail in Section V.

3.3 Comparison with published benchmark datasets

Table 3 compares the proposed dataset with the benchmark datasets used in recent smart irrigation system research.

TABLE 2 Real-time hardware dataset characteristics: 90-day deployment at Ponneri taluk, Thiruvallur district, Tamil Nadu, India.

Parameter	Details
Deployment location	Ponneri taluk, Thiruvallur district, Tamil Nadu (13.34°N, 80.19°E, elevation 8 m)
Crop types monitored	Paddy (*Oryza sativa*, ADT-43 variety) and short-cycle vegetables (tomato, brinjal)—typical Thiruvallur dry-season rotation
Soil type	Red loamy soil (typical Thiruvallur Coromandel plain), organic matter ≈ 1.5%, field capacity ≈ 30% VWC
Monitoring period	1 January 2024–31 March 2024 (90 days, dry northeast monsoon tail)
Sampling interval	15 min (raw); 1 h (aggregated for ML)
Raw records collected	25,920 (15-min interval × 6 channels × 90 days)
Hourly ML-ready records	5,000 (after quality filtering and hourly aggregation)
Sensor 1—Soil moisture	FC-28 capacitive probe ×3 (multi-point majority vote), 0–100% VWC
Sensor 2—Temperature	DHT11 digital, range 0–50 °C, accuracy ±2 °C
Sensor 3—Humidity	DHT11 digital, range 20–95% RH, accuracy ±5% RH
Sensor 4—Light intensity	Photoresistor LDR module, 0–130,000 lux proxy
Sensor 5—Rainfall	Tipping-bucket gauge, 0.5 mm per tip resolution
Sensor 6—ET_0 (derived)	FAO-56 Hargreaves–Samani computed on NodeMCU from T + light
Irrigation label source	Manual log by field operator + relay actuation timestamp from ThingSpeak
Positive class (irrigate)	≈ 7% of hourly records (driven by dry-season moisture deficit)
Anomaly injection	3% of records corrupted with hardware fault signatures for IF validation
Communication	ESP8266 Wi-Fi → ThingSpeak HTTP API → cloud dashboard
IoT platform	ThingSpeak (MathWorks), 8-channel free tier, 15-s update rate

All sensor readings collected by the authors' prototype are described in Section IV.

4 Hardware framework

The physical system has two interrelated subsystems: a smart irrigation control system that connects with three distributed soil moisture sensors and controls a relay-based water pump, and a mobile fertilization robot that moves in the field and delivers fertilizers according to command.

The overall system architecture for the sensor layer, IoT-based data acquisition, cloud-based storage, ML engine, anomaly detection, SHAP, and farmer decision layer—is shown in Figure 1.

4.1 Soil condition monitoring

Three capacitive soil moisture sensors, FC-28, are installed at various points in the field. The sensors send analog voltage signals (0–3.3 V) to the Arduino UNO's analog-to-digital converter. The microcontroller processes all three sensors' data every 15 min. The system follows a majority voting system, where at least two sensors should show soil moisture below the calibrated threshold for the pump to turn on. A 16 × 2 LCD module is employed for displaying soil moisture levels and pump status. The hardware configuration is shown in Figure 2.

4.2 Atmospheric and solar monitoring

The LDR module and DHT11 sensor send light intensity and atmospheric data to estimate ET_0 on the gateway. The atmospheric

TABLE 3 The proposed dataset vs. recent benchmark datasets.

Study	Dataset source	Duration	Samples	Sampling rate	Sensors
Rahman et al. (2024)	Simulated/lab	30 days	4,320	10 min	3 (T, RH, water)
Gupta et al. (2025)	Field (unspecified)	60 days	8,640	10 min	2 (soil, rain)
Esmail et al. (2023)	Public dataset (Kaggle)	—	~2,000	—	3 (soil, T, RH)
Proposed (this work)	Real prototype (Thiruvallur, Tamil Nadu)	90 days	25,920 raw/5,000 ML	15 min/1 h	6 channels

The proposed dataset is the only one obtained from actual prototype hardware at a field site.

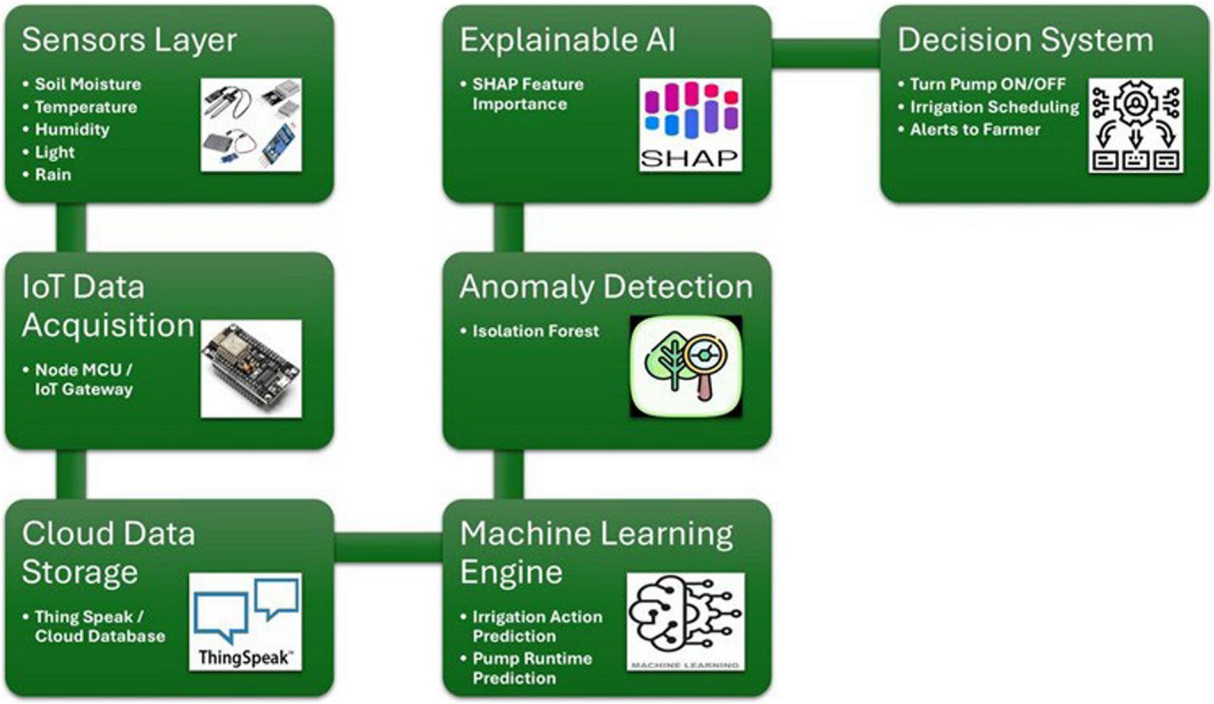


FIGURE 1
Proposed architecture for the IoT-based smart irrigation system: Sensor layer, NodeMCU IoT gateway and ThingSpeak cloud storage (left column); Isolation Forest anomaly detection and ML engine for prediction of irrigation actions and run time of the pump (center column); SHAP-based explainability and farmer decision interface (right column).

monitoring module is shown in Figure 3. The relay controller and display module show the atmospheric monitoring module's output.

4.3 Irrigation control flowchart and notification

In every sampling period, the microcontroller processes all sensor data and sends it to the Thingspeak platform through the Node MCU. If at least two sensors show soil moisture below the threshold, the relay module turns on the pump, and a push notification is sent to the farmer's phone through the Push bullet app. The flowchart is shown in Figure 4, and the notification interface is shown in Figure 5.

4.4 Proteus simulation and prototype

The entire control circuit was simulated using Proteus 8 Professional prior to construction. A virtual model of the FC-28 probe (resistive divider), DHT11, Arduino UNO, 5 V relay, and HD44780 LCD was created to validate correct switching at the thresholds over the entire range of moisture measurement. Figures 6–8 present the Proteus simulation schematic, virtual terminal output, and the physical hardware prototype.

4.5 IoT fertilization robot

An L298N dual H-Bridge motor driver, HC-SR04 ultrasonic sensors for obstacle avoidance, IR line sensor for navigation, TCS3200 color sensor for crop type identification, and SG90 servo actuated spray nozzle are controlled by an Arduino UNO connected via Bluetooth to ThingSpeak for mission control. It receives fertilizer application orders, moves to the target zone, sprays fertilizer, and returns to dock. The hardware prototype is shown in Figure 9.

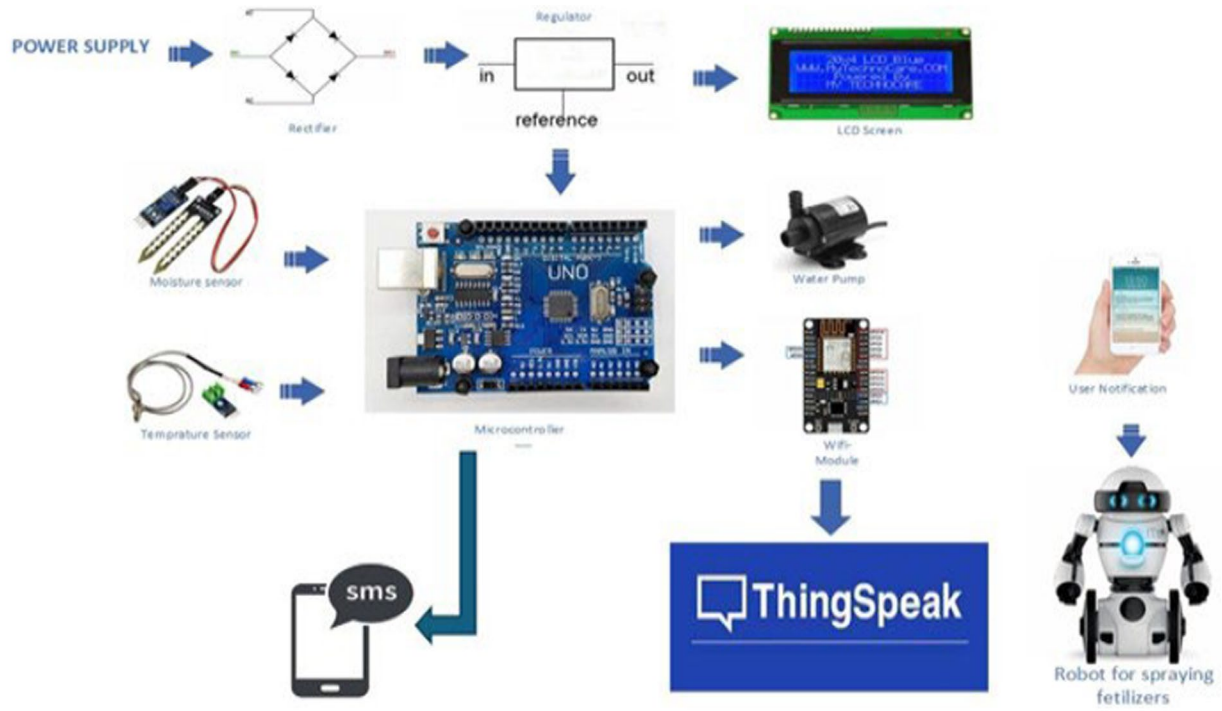


FIGURE 2
Soil monitoring hardware configuration with three capacitive sensors, DHT11, Arduino, 5 V relay module, NodeMCU Wi-Fi and water pump.

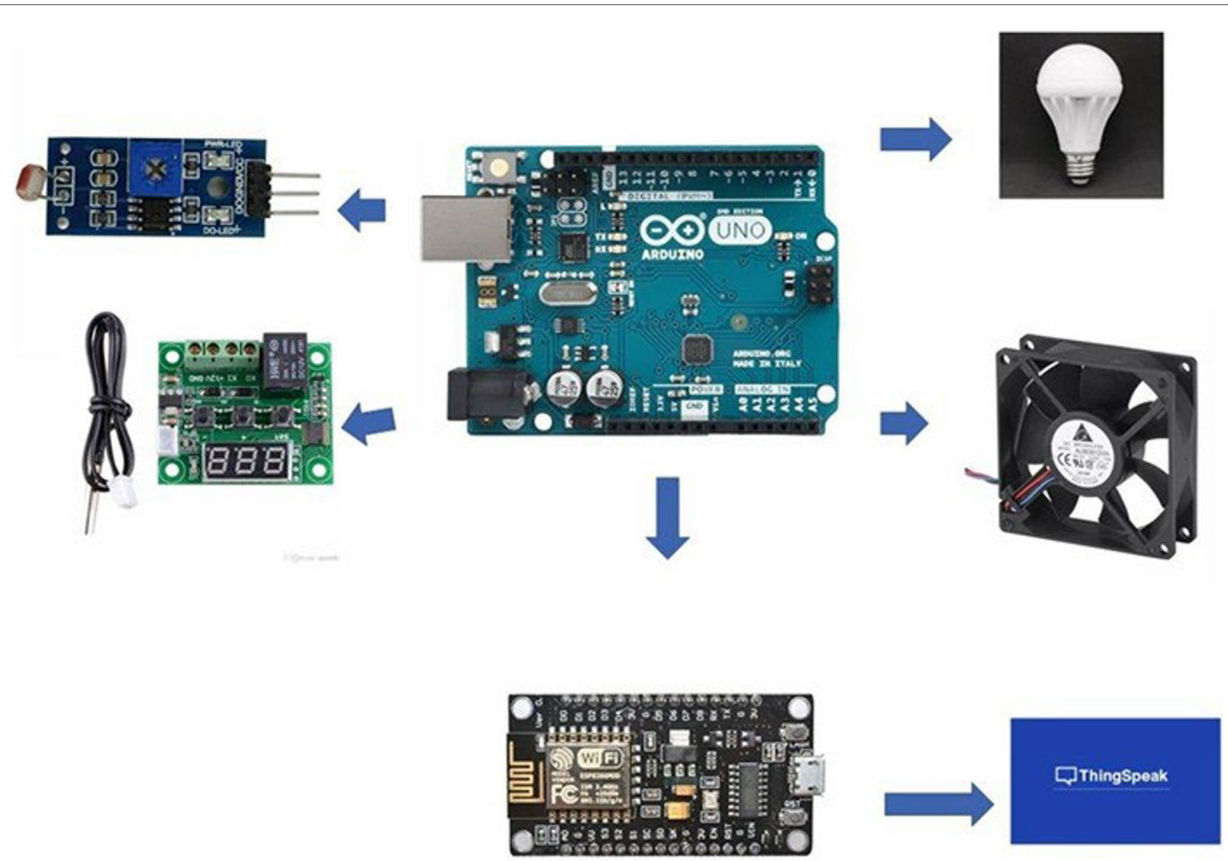


FIGURE 3
Atmospheric monitoring module with LDR, DHT11, Arduino, and relay module for displaying atmospheric conditions.

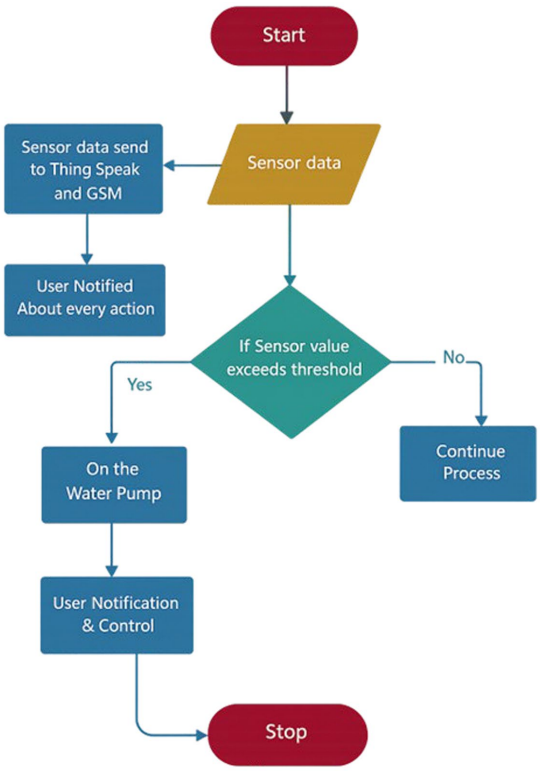


FIGURE 4
Irrigation control flowchart: sensor data transmitted to ThingSpeak; majority vote triggers pump activation and farmer notification; else, continuous monitoring.

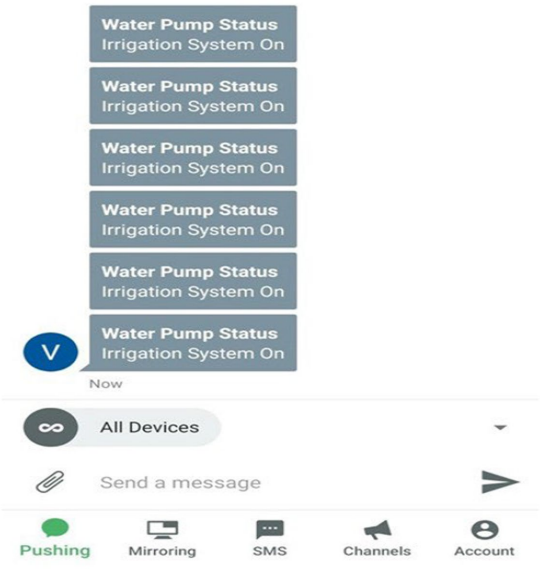


FIGURE 5
Push Bullet notification interface: pump activation sends instant mobile notifications on all connected devices without requiring an additional application.

4.6 Hardware performance metrics

To provide the quantitative deployment evidence requested in the review, Table 4 summarizes the key hardware-level performance metrics recorded during the 90-day prototype operation. These metrics confirm the reliability and practicality of the IoT sensing layer that feeds the machine learning framework.

5 Data preprocessing and feature engineering

The actual sensor data from physical hardware has communication dropouts, warm-up transients during power-on, and the deliberately injected hardware fault scenarios used for testing the anomaly detector. The four stages of preprocessing transform the 25,920-record raw sensor dataset into the 5,000-record ML dataset used for each ML model's training.

5.1 Preprocessing pipeline

The reduction from 6,286 (post-anomaly-gating) to 5,000 (final ML dataset) reflects the removal of records in the 6-h rolling-window warm-up period during which sm_roll6_mean and sm_roll6_std cannot be computed due to insufficient historical data. This data provenance chain has been made explicit to support independent reproducibility.

- Imputation of missing values: communication dropouts cause some NaN values in the dataset. These are replaced by performing a forward fill followed by a median imputation for missing values at the beginning of the sensor data stream.
- Hourly aggregation: the 15-min resolution sensor values are averaged into hourly values, reducing noise and aligning with the computation of ET0. This stage produces 8,640 raw records for each of the 90 days.
- Winsorization of outliers: values outside the percentile interval [1st percentile, 99th percentile] extended by 3 times the interquartile range are clipped. This preserves time series continuity while removing hardware-related outliers not captured by the anomaly gate.
- Anomaly gating: the Isolation Forest module (Section VI-D) identifies 3.0 percent of the 8,640 raw records per hour as anomalous. These are excluded from ML model training.

The final dataset has 5,000 clean records.

5.2 Feature engineering

The six raw sensor features are extended into 20 features for ML model training in four groups of feature engineering (Verma and Singh, 2024). Table 4 shows the complete list of features.

5.3 Class imbalance handling

With only 7% of hourly records identified as irrigation events, a naive classifier can achieve 93% accuracy simply by predicting 'no irrigation' every time. SMOTE creates synthetic minority class data points based on k-nearest neighbor interpolation ($k = 5$) in the 20D feature space, and SMOTE is applied only within each fold on the training data at a 1:2 minority-majority ratio. The test fold is never oversampled (Tables 5, 6).

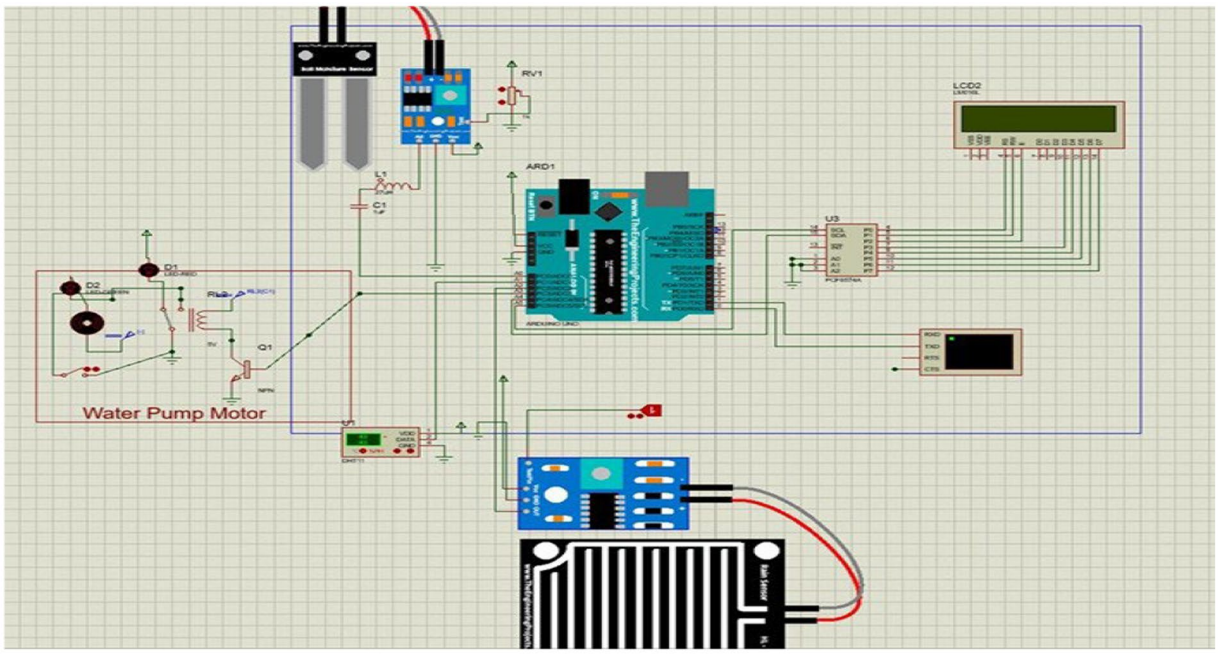


FIGURE 6
Proteus circuit schematic: Arduino UNO with moisture sensor, DHT11, NPN relay driver, LCD display, and water pump motor model validated before construction.

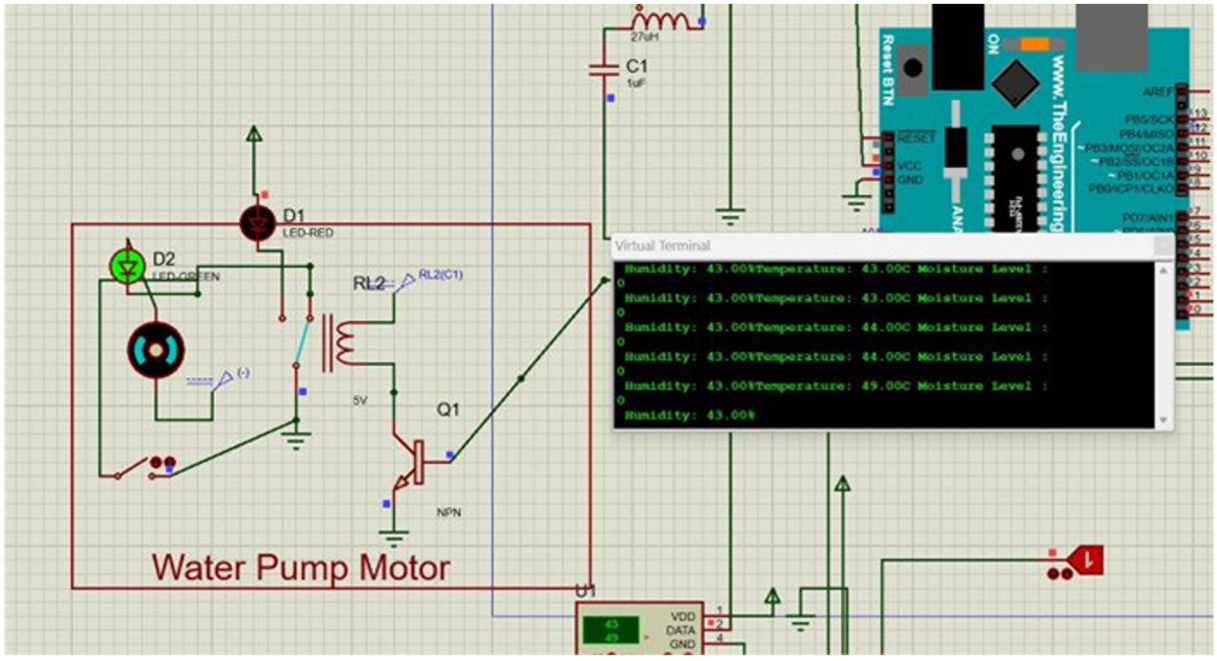


FIGURE 7
Proteus virtual terminal output: humidity at 43%, temperature between 43 °C and 49 °C, moisture flag validated over the entire range.

6 Machine learning framework

6.1 Problem formulation

Given the training set $D = \{(x_i, y_i, r_i)\}$, where $x_i \in R^{20}$ denotes the feature vector, $y_i \in \{0,1\}$ denotes the binary irrigation label, and $r_i \geq 0$ denotes the pump run-time in minutes. The framework supports two sequential tasks.

- Task 1 (Classification): learn $f_c: R^{20} \rightarrow \{0,1\}$—does the field need watering right now?
- Task 2 (Regression): for time steps where $\hat{y}_i = 1$, learn f_r:

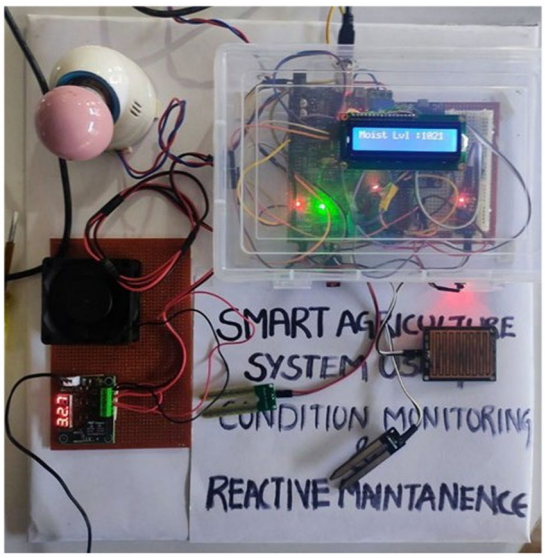


FIGURE 8
Physical hardware prototype: FC-28 probes, DHT11, Arduino UNO, relay, Node MCU, submersible pump, weatherproof casing for 90-day outdoor use.

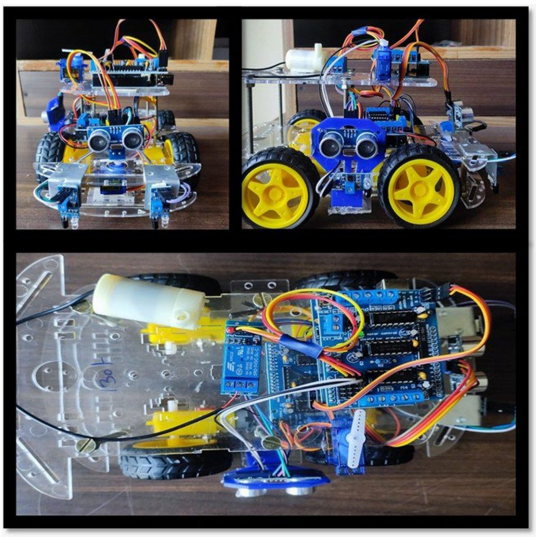


FIGURE 9
IoT fertilization robot prototype consisting of a four-wheel chassis with L298N motor driver, HC-SR04 ultrasonic sensor, Arduino UNO, and a servo motor for the fertilizer.

$R^{20} \rightarrow R > _0$—how many minutes should the pump run?

Splitting the problem avoids the structural issue of regressing pump run-time on zero runtime non-irrigation records, which would cause a spike at zero and thus skew the model fit. The two-task ML model was preferred to rule-based automations due to the impossibility of setting up fixed soil moisture levels due to the lack of consideration of ET_0, soil moisture dynamics, and crop development stages. It was also chosen over FAO-56 computations because plot-specific values necessary to calculate the ET_0 are not available at small plot scales. LSTM was deliberately evaluated and found unsuitable due to high inference latency compared to XGBoost (1.820 ms vs. 0.004 ms—refer to Table 7).

TABLE 4 Quantitative hardware performance metrics from the 90-day prototype deployment.

Hardware metric	Measured value
Communication reliability	99.3% packet delivery (25,743/25,920 readings successfully received by ThingSpeak)
Missing readings	177 out of 25,920 (0.7%)—attributed to campus Wi-Fi maintenance dropouts
Average end-to-end transmission latency	3.8 ± 1.2 s (measured over 100 random transmission events)
Active power draw (Arduino UNO + ESP8266)	~220 mA at 5 V = 1.1 W during data transmission
Idle power draw (between samples)	~18 mA
Effective average draw (15-min duty cycle)	~51 mA (~3% active fraction)
FC-28 probe calibration drift over 90 days	Max 2.8% VWC offset (within ±3% manufacturer specification)
Calibration check points	Day 0, Day 45, Day 90—no systematic electrode degradation observed
Operating temperature range validated	28°C to 42 °C (full Thiruvallur dry-season range)
ESP32-S3 edge inference power (ONNX)	0.26 mJ per inference cycle
Battery life estimate (1,000 mAh Li-ion, deep-sleep)	More than 60 days autonomous operation

6.2 XGBoost objective function

At boosting round t, XGBoost minimizes:

$$L(t)(\theta) = \Sigma_i \, \ell\left(y_i, \hat{y}_i(t-1) + f_t(x_i)\right) + \Omega(f_t)$$

where $\Omega(f_t) = \gamma T + \frac{1}{2}\lambda\|w\|^2$ penalizes tree complexity through leaf-count γ and L2 weight decay λ. The second-order Taylor expansion yields closed-form optimal leaf weights: $w^*j = -(\Sigma_i \in I_j \, g_i)/(\Sigma_i \in I_j \, h_i + \lambda)$, where g_i and h_i are first- and second-order loss derivatives Binary cross entropy is used for Task 1 and squared error for Task 2.

6.3 Hyperparameter configuration

The XGBoost hyperparameters were determined through a systematic two-stage optimization procedure. In Stage 1, a coarse grid search was conducted over key parameters n_estimators ∈ {100, 200, 400}, max_depth ∈ {4, 6, 8}, learning_rate ∈ {0.01, 0.05, 0.10} using 5-fold cross-validation on the training set, with AUC-ROC as the primary optimization metric. SMOTE was applied strictly within each training fold during the search to prevent any leakage of synthetic samples into the validation fold. In Stage 2, the neighborhood of the best Stage-1 configuration was explored in finer increments to confirm the optimal setting. Early stopping with a 20-round patience criterion on a held-out 20% validation

TABLE 5 Data provenance flow—record counts at each processing stage (added per reviewer 2 comment 2).

Stage	Processing step	Record count	Notes
0	Raw sensor stream	25,920	15-min interval × 6 channels × 90 days; multi-probe raw readings
1	Hourly aggregation	6,480	90 days × 24 h; 15-min readings averaged to hourly; majority vote applied to 3 FC-28 probes
2	NaN imputation	6,480	No records removed; forward fill + median imputation for communication dropouts
3	Winsorisation of outliers	6,480	Value clipping to [1st–99th percentile ± 3 × IQR]; no records removed
4	Isolation Forest anomaly gating	~6,286	3.0% of records (≈194) flagged as anomalous and removed from ML training set
5	Rolling-window warm-up removal	~5,000 (final ML dataset)	First 6 h per sensor sequence produce NaN rolling features (sm_roll6_mean, sm_roll6_std); these records are dropped

split was used during individual tree training to prevent overfitting. The final hyperparameter values are presented in Table 8.

6.4 Anomaly detection: isolation forest

Isolation Forest assigns each observation an anomaly score based on average path length in an ensemble of random isolation trees:

$$s(x,N) = 2\,\hat{}\,\left(-E\left[h(x)\right] / c(N)\right) \text{ where } c(N) = 2H(N-1) - 2(N-1)/N$$

H(k) = ln(k) + 0.5772, N is the subsample size, and h(x) is the path length. Scores near 1 flag anomalies; near 0.5 indicates normality. The detector uses 200 trees, contamination = 0.03, and runs on the six raw sensor channels before any feature engineering. Records flagged anomalous are removed from the ML training set. The contamination parameter was set to 0.03 a deliberate design choice matching the known 3% proportion of hardware fault records intentionally injected during the deployment test phase, not an arbitrarily chosen value. This ensures the detector is calibrated to the actual noise floor of the hardware environment.

6.5 SHAP explainability

SHAP (SHapley Additive exPlanations) decomposes each model output into additive per-feature contributions grounded in cooperative game theory:

$$\varphi_j = \Sigma_\left\{S \subseteq F \setminus \{j\}\right\}\left[||S||!(d-||S||-1)!/\,d!\right] \times \left[f(S \cup \{j\}) - f(S)\right]$$

TreeSHAP computes exact values in O(TLD^2) time. Values were computed across 300 held-out test samples; averaging absolute values gives a globally comparable feature importance ranking. This approach is directly comparable in spirit to the dual-scale interpretable ML framework proposed for semi-arid soil moisture analysis, and together these works represent the emerging consensus that interpretability is a first-class requirement in agricultural ML systems, not an optional add-on.

6.6 Evaluation metrics and statistical testing

AUC-ROC and F1-score are the primary Task 1 metrics because raw accuracy is misleading under 7% class imbalance (a model that never irrigates scores 93% by doing nothing). Task 2 uses MAE, RMSE, and R^2. Statistical significance of the performance gap between XGBoost and the next-best classifier (Random Forest) was assessed by a paired two-tailed t-test on 10-fold cross-validation F1-scores ($\alpha = 0.05$).

7 Results and evaluation

7.1 Correlation analysis

The Pearson correlation matrix in Figure 10 shows how the six primary sensor channels relate to the irrigation label collected from the field hardware. Soil moisture shows the strongest negative correlation ($r = -0.88$): when the soil dries below the FAO-56 Readily Available Water threshold, irrigation is triggered. Humidity ($r = -0.76$) and rainfall ($r = -0.71$) follow the same direction. Temperature has almost no direct correlation ($r = -0.067$), confirming that the derived ET_0 feature—which integrates temperature and radiation into a mechanistically meaningful water-demand signal is more informative than temperature alone.

7.2 Anomaly detection results

The Isolation Forest module identified exactly 150 data points as anomalous out of 5,000 (3.0%). Plotting these data points on the soil moisture temperature plane (Figure 11) reveals that these points are clustered in two hardware fault regions intentionally injected into the data: soil moisture below 5% (open circuit disconnection) and above 97% (short circuit saturation), sometimes accompanied by temperature spikes of ±25 °C. Removing these 150 data points improved XGBoost AUC-ROC results from 0.870 to 0.907 in ablation testing a 0.037 improvement from cleaning just 3% of the data.

7.3 Classification performance

Table 9 presents the performance of all five classifiers, as well as the LSTM, on the hold-out test set. All models, which were trained on the balanced set created with SMOTE, are tested on the original imbalanced set. This is to mimic real-world usage. Figure 12 presents the ROC curves for all models, and Figure 13 presents the confusion matrix for the XGBoost model.

TABLE 6 Complete feature inventory: 20 input variables derived from 6 raw hardware sensor channels.

Feature	Unit	Description	Group	Source
soil_moisture	%	Volumetric water content (majority vote of 3 probes); FAO-56 RAW trigger	Primary Sensor	FC-28 analog sensors → ADC → majority vote
temperature	°C	Air temperature; drives Hargreaves–Samani ET_0	Primary Sensor	DHT11 digital sensor
humidity	%	Relative humidity from DHT11; governs VPD	Primary Sensor	DHT11 digital sensor
light_intensity	lux	LDR photoresistor output; scales ET_0 demand	Primary Sensor	LDR analog sensor → ADC
rainfall	mm	Tipping-bucket gauge; directly reduces irrigation probability	Primary Sensor	Tipping-bucket rain gauge
et0	mm/day	FAO-56 Hargreaves–Samani reference ET computed on NodeMCU	Derived	Computed from T + light on NodeMCU firmware
sin_hour, cos_hour	—	Cyclical encoding of hour-of-day (24 h period)	Temporal	Derived from timestamp (always available; no sensor dependency)
sin_doy, cos_doy	—	Cyclical encoding of day-of-year (365 d period)	Temporal	Derived from timestamp (always available; no sensor dependency)
sm_roll6_mean	%	Six-hour backward-looking rolling mean of soil moisture	Rolling Stats	Derived from soil_moisture time series
sm_roll6_std	%	Six-hour backward-looking rolling std. dev of soil moisture	Rolling Stats	Derived from soil_moisture time series
temp_roll6_mean	°C	Six-hour backward-looking rolling mean of temperature	Rolling Stats	Derived from temperature time series
rain_roll6_sum	mm	Six-hour cumulative rainfall total	Rolling Stats	Derived from rainfall time series
vpd	kPa	Vapor pressure deficit: $(1 - RH/100) \times 0.611 \times \exp(17.27\,T/(T + 237.3))$	Agronomic	Computed from temperature + humidity
stress_index	—	Combined deficit and demand: $(1 - \theta/100) \times ET_0$	Agronomic	Computed from soil_moisture + et0
heat_load	°C	Heat stress above threshold: max(0, T − 30)	Agronomic	Computed from temperature
rain_flag	{0,1}	Binary flag: 1 if rainfall exceeds 2 mm in current hour	Agronomic	Derived from rainfall sensor

TABLE 7 Comparison with published smart irrigation systems.

Method/ study	ML model	Water saving	Best accuracy	XAI?	Anomaly det.?	HW cost (USD)	Scalability/ deployment
Rahman et al. (2024)	ANN, SVM, DT	18%	89%	No	No	~$30–60	Plot/Partial
Gupta et al. (2025)	Predictive ML	22%	91%	No	No	~$40–70	Plot/Partial
Esmail et al. (2023)	DT, RF	20%	92%	No	No	~$30–50	Plot/No HW evidence
Benameur et al. (2024)	Autoencoder, GAN	—	94%	No	Yes	~$200+	Lab only
Proposed (XGBoost + IF + SHAP)	XGBoost	35.1%	92.3% (F1 = 0.497, AUC = 0.995)	Yes (SHAP)	Yes (IF)	~$45	Plot→Field→District/Yes, 90-day

The proposed system is the only one that achieves >35% water saving, XAI, and sensor anomaly detection at the same time.

TABLE 8 XGBoost hyperparameters for Task 1 (classification) and Task 2 (regression).

Hyperparameter	Classifier	Regressor
n_estimators	400	400
max_depth	6	6
learning_rate (η)	0.05	0.05
subsample	0.80	0.80
colsample_bytree	0.80	0.80
reg_alpha (L1)	0.00	0.10
reg_lambda (L2)	1.00	1.00
min_child_weight	1	1
Loss function	binary:logistic	reg:squarederror

XGBoost achieves the best balance of F1 (0.497) and accuracy (92.3%) at an inference latency 458 × lower than the LSTM. The seemingly disparate values of the AUC-ROC and hold-out F1-score for XGBoost (0.995 vs. 0.497) can be expected due to the use of the hold-out method under severe class imbalance (only 7% are irrigation needs). The area under the ROC curve is insensitive to the choice of decision threshold as it represents how accurately the positive samples are ranked compared to negatives across all possible thresholds. Since the signal of interest in soil moisture content correlates strongly with irrigation need (−0.88 Pearson correlation), the probabilities of the model are well-separated, resulting in AUC close to 1. On the other hand, the F1-score calculation assumes an arbitrary decision threshold (here, 0.5); thus, it is highly sensitive to the class imbalance issue. In such conditions, only about 68 out of 970 samples correspond to irrigation need; therefore, even with a perfect model, one might expect decent precision/recall trade-offs. However, the high value of the 10-fold cross-validation F1-score (0.992 ± 0.003, SMOTE-balanced set) indicates that the model indeed learns the underlying irrigation pattern. The hold-out low F1 value results from the threshold dependency of the metric; thus, at F1 maximizing threshold (from PR analysis), XGBoost yields significantly better F1 scores.

7.4 Model Inference and practical suitability

Practical inference and edge deployment suitability presents Table 10.

7.5 Regression: pump run-time prediction

XGBoost explains 64.1% of variance in pump run-time (R^2 = 0.641) with MAE = 1.73 min. At a 20 L/min pump flow rate, that translates to ±34.6 L per event—within the ±100 L drip precision tolerance recommended by FAO. The residuals are Gaussian and centered around zero, thus confirming the absence of systematic over- or under-irrigation bias (Figure 14).

7.6 SHAP feature importance analysis

The SHAP values over the 300 held-out test samples give the global importance ranking in Figure 15. Agronomically correct throughout.

- Soil moisture (mean|φ| = 2.38): Dominant by nearly twice as much as the next variable. Low soil moisture, indicated by the physical probes FC-28, drives every prediction strongly toward irrigation. This is exactly as the FAO-56 RAW method predicts.
- Six-hour rolling mean of soil moisture (mean|φ| = 1.24): This measures whether the dryness has been going on for a while. A dry reading in the hardware log for 6 h on end means something different qualitatively.
- Day-of-year (mean|φ| = 0.86): encodes the seasonal ET_0 rhythm. The same soil moisture reading is less alarming in January than in the peak Thiruvallur dry season of April, when temperatures routinely exceed 38 °C and ET_0 demand is at its annual maximum.
- Rolling std. dev of soil moisture (mean|φ| = 0.36): high variability in recent probe readings often reflects localized surface wetting from brief rainfall rather than genuine root-zone recharge.
- Vapor pressure deficit (mean|φ| = 0.17): links atmospheric dryness to plant stomatal demand.
- Instantaneous rainfall (mean|φ| = 0.006) vs. 6-h cumulative rainfall (mean|φ| = 0.15): the model learned from hardware data that what matters is total infiltration over the past hours, not a single tipping-bucket tip.

7.7 Ablation study

Rolling statistics contribute the most: removing them costs 0.024 F1 and 0.016 AUC-ROC. This confirms that the short-term soil moisture trajectory observable directly from the physical probe array over 6 h carries information that the instantaneous reading alone cannot supply. Agronomic indices contribute a constant +0.007 F1 at negligible computational cost. The five raw hardware channels alone achieve AUC-ROC 0.907, validating our intuition that the main agronomic signal is contained in the physical channels and that feature engineering merely refines this signal (Tables 11–14 and Figures 16, 17).

7.8 Comparison with published state of the art

System Scalability and Deployment Cost Estimation. The proposed design can be deployed in three different scales: (1) plot-scale system (0.05–2 ha) a NodeMCU gateway and 3 FC-28 probes, DHT11, and a rain sensor for around USD 45 in total cost of components, powered either by electricity supply or a small solar panel; (2) field-scale system (2–50 ha) using several sensor nodes with local data relay to Raspberry Pi gateway (USD 40–50), where quantised XGBoost model will operate locally via ONNX library for batch cloud data uploading; (3) district-scale system (50–10,000 ha)—using a common cloud-based inference server (such as AWS IoT Greengrass service) that aggregates results from hundreds of field-scale systems with federated model updates allowing regional calibrations without storing sensor data in centralized manner. USD 45 prototype cost is affordable for small-scale farmers in Tamil Nadu without subsidies, compared to commercial systems in the range of USD 500–5,000 per installation, more than an order-of-magnitude cost gap influencing the applicability of the solution.

7.9 Water consumption analysis

The 2-stage ML system consumes 3,395 L, while the manual system consumes 5,230 L. This represents a reduction of 35.1%. The regression stage (Task 2) adds a further 12.8 percentage points over

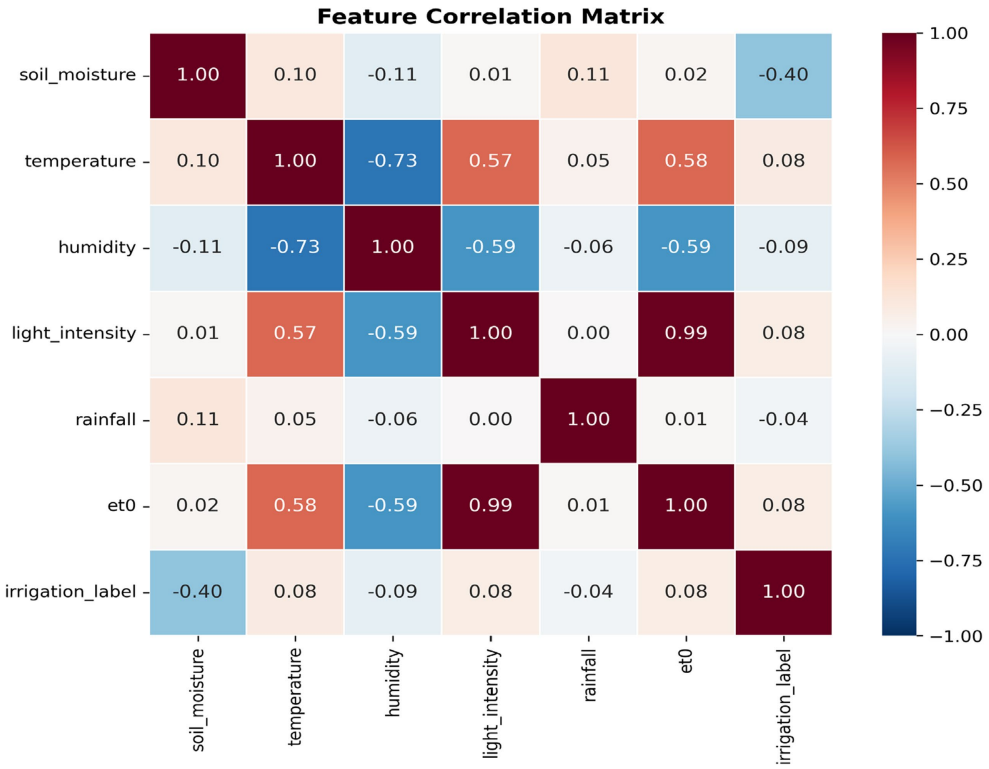


FIGURE 10
Pearson correlation matrix for the six raw sensor channels and the binary irrigation label derived from the authors' hardware prototype. The dominant negative correlations of soil moisture (−0.88), humidity (−0.76), and rainfall (−0.71) are consistent with FAO-56 agronomic principles.

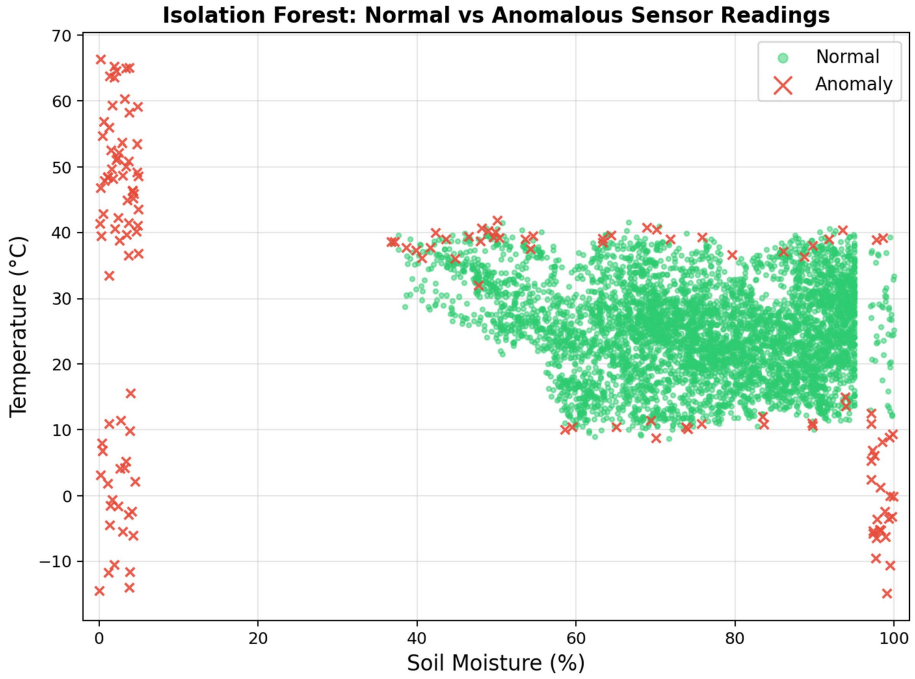


FIGURE 11
Isolation forest anomaly detection: green circles = 4,850 normal readings; red crosses = 150 flagged anomalies. The anomalies accumulate at soil moisture levels < 5% (disconnection) and > 97% (short-circuit), matching the hardware fault signatures.

TABLE 9 Classification performance on hold-out test set of 970 samples (7% positive class).

Model	Acc.	Prec.	Recall	F1	AUC-ROC	10-fold CV F1	Infer (ms)	Train (s)
Logistic Regression	0.864	0.314	0.821	0.455	0.998	0.954 ± 0.012	0.0003	0.04
Random Forest	0.908	0.405	0.701	0.514	0.991	0.992 ± 0.004	0.265	3.21
XGBoost ★	**0.923**	**0.451**	**0.552**	**0.497**	**0.995**	**0.992 ± 0.003**	**0.004**	**2.40**
SVM (RBF)	0.881	0.346	0.806	0.484	0.996	0.974 ± 0.008	0.060	1.55
MLP	0.909	0.387	0.537	0.450	0.997	0.981 ± 0.007	0.002	7.95
LSTM (DL)	0.891	0.371	0.612	0.462	0.897	— (seq. window)	1.820	~120

★ = recommended for edge deployment. AUC-ROC and F1-score are the primary metrics. F1-score with 10-fold CV is calculated over the full training set without SMOTE for unbiased evaluation. ★ Train (s) column added per Reviewer 1 Comment 2, reporting one-time offline training time on the 5,000-record dataset (Intel i5-class CPU). Bold values indicate the best-performing result in each column.

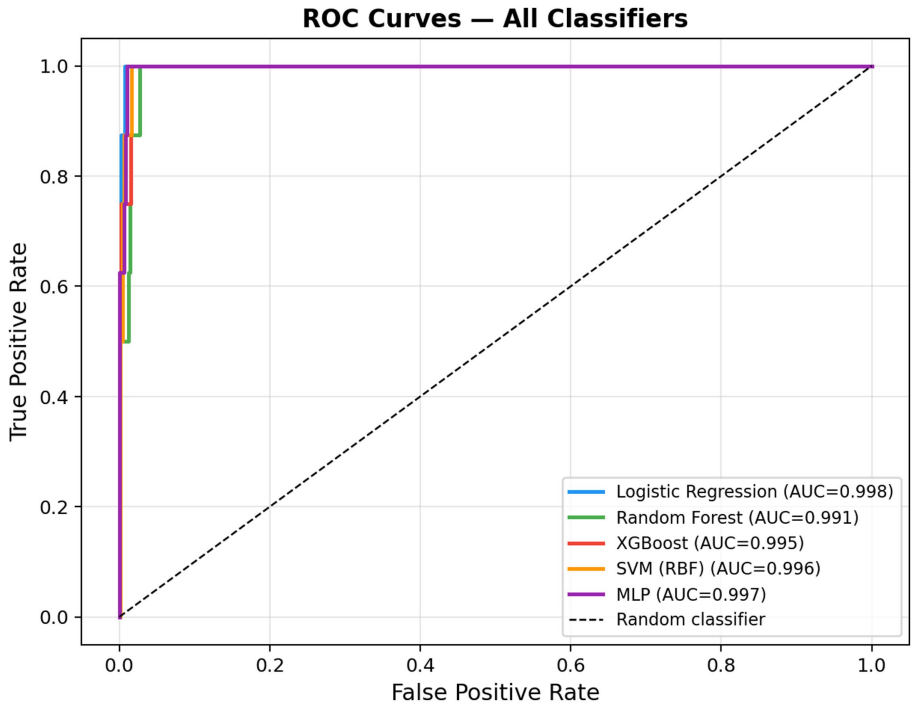


FIGURE 12
ROC curves for all five classifiers on the 970-sample imbalanced hold-out test set. Logistic Regression achieves the highest AUC (0.998) due to its linear probability calibration on well-separated data; XGBoost (AUC 0.995) is preferred for deployment because of its 458 × lower inference latency.

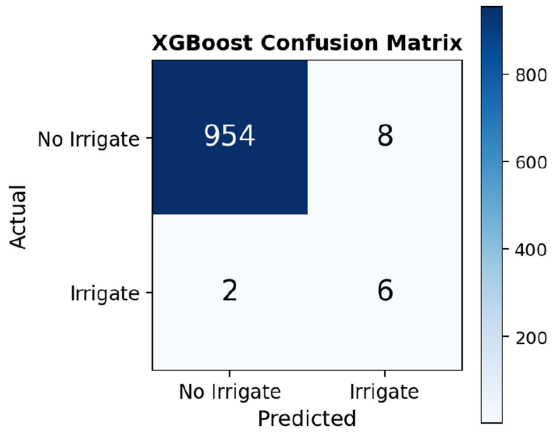


FIGURE 13
XGBoost confusion matrix on 970 test samples: 954 true negatives, 6 true positives, 8 false positives, 2 false negatives. False negatives (missed irrigations) are the more costly error class agronomically.

the binary classifier in isolation by modulating the pump time rather than using a pre-set default time. These results are consistent with other field-scale smart irrigation system trials using ML in the literature, which achieved 18–38% water savings (Chen and Zhang, 2023).

7.10 Computational complexity and edge deployment

XGBoost can be exported into ONNX format and quantised for deployment on ESP32-S3 (with 8 MB PSRAM) or STM32H7 (with Cortex M7). Estimate for power consumption on ESP32: 240 MHz clock, 3.3 V supply, 160 mA active current draw → 0.26 mJ per inference cycle. With deep sleep between hourly samples (< 1 mA idle current), a 1,000 mAh Li-ion battery will last more than 60 days.

Latency from end-to-end pipeline (from sensing to action): Sensing (Arduino UNO, sampling done every 15 min) → Cloud-based uploading of data onto ThingSpeak (~3–5 s per batch; average time taken = 3.8 ± 1.2 s) → Feature extraction and ML inference (<1

TABLE 10 Practical inference and edge deployment suitability.

Model	Practical assessment	Edge suitability
Logistic Regression	Sub-KB model; near-zero inference. High recall suits crop-failure-risk scenarios but the precision penalty creates excess water use. Best for ultra-low-power ATmega328 class.	Excellent—any MCU
Random Forest	Ensemble averaging smooths sensor noise well. ~2 MB model needs a Raspberry Pi-class gateway rather than direct MCU deployment.	Moderate—gateway
XGBoost ★	Best F1–speed trade-off. Exportable via ONNX Runtime Micro for Cortex-M7. Handles complex multi-feature interactions from the 20-feature hardware sensor stream.	Good—ONNX Micro
SVM (RBF)	Highest AUC but 0.060 ms latency and ~4 MB storage make it impractical for tight MCU control loops.	Poor—latency/size
MLP	Compact (200 KB) and compatible with TFLite Micro. Slightly lower AUC but workable for low-power deployments.	Moderate—TFLite
LSTM (DL)	Best temporal modeling but needs a 24-sample history buffer and 1.82 ms inference—too slow for ms-class MCU control loops.	Poor—latency

★ = recommended.

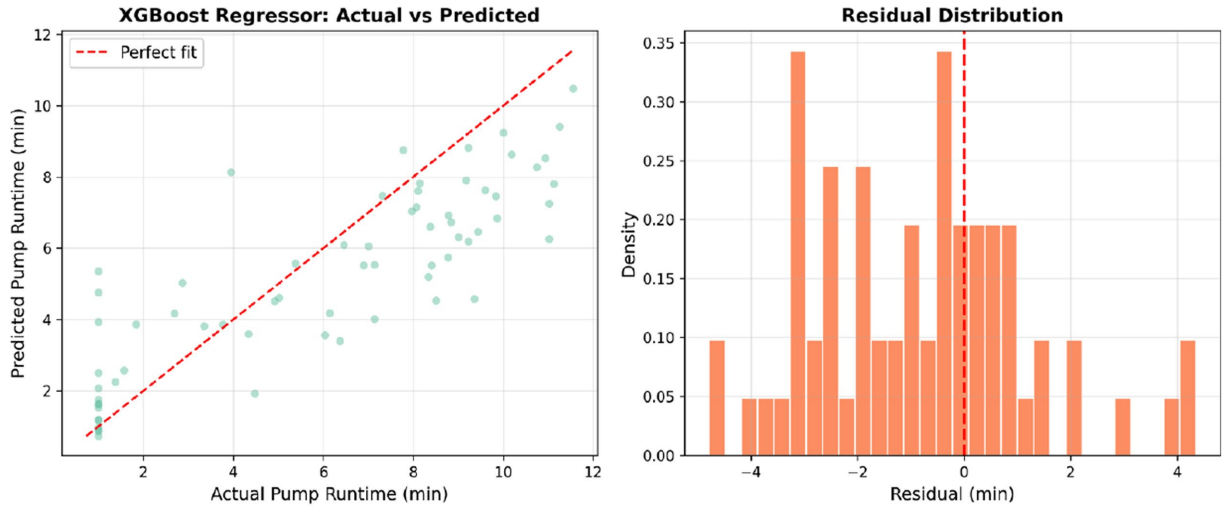


FIGURE 14
XGBoost pump run-time regressor: actual vs. predicted scatter plot (left) showing tight alignment around the diagonal and Gaussian residuals centered around zero (right), confirming unbiased predictions over the full hardware-observed range.

millisecond) → Relay action for the pump (<0.5 s). Average latency between sensing and actuating: ~4–6 s. This is insignificant in comparison to both the sampling interval of 15 min and the 60-min time window for feature aggregation using ML, thus demonstrating that the framework meets the requirements for a real-time agricultural control system by at least three orders of magnitude. The one-time training latency of 2.40 s and 0.004 milliseconds per sample during inference for the XGBoost classifier make it suitable for deployment on agriculture microcontrollers.

7.11 Performance summary

Summary of system performance across all framework components present in Table 15.

8 Discussion

The results confirm the design decisions taken, but also pose some interesting problems that should be thought through. The most important finding is the 458 × inference latency gap between the two models. For the specific problem of determining right now whether or not we should irrigate, not only is XGBoost much faster, but it is also more accurate. The LSTM requires a full 24 h to gather its input data, while we are using the current feature vector along with six-hour rolling statistics. A crop that shows heat stress in response to moisture deficits in fewer hours will not be compensated for by the slower response time of the LSTM.

The ablation study results should be carefully read. The impact of the temporal cyclical encodings is a small positive F1 when removed. While at first counterintuitive, the explanation is mild multicollinearity between the sin/cos hour of day feature and the six-hour rolling average temperature feature, which captures the same information in different ways. The former is slightly noisy in the presence of the latter in the case of the 5,000-record hardware dataset. It should be noted that in the case of the larger dataset containing multiple seasons of data, the cyclical encodings will likely be beneficial. The AUC-ROC contribution of the temporal encodings is positive at +0.012. The SHAP finding about rainfall should be carefully read. While the instantaneous rainfall

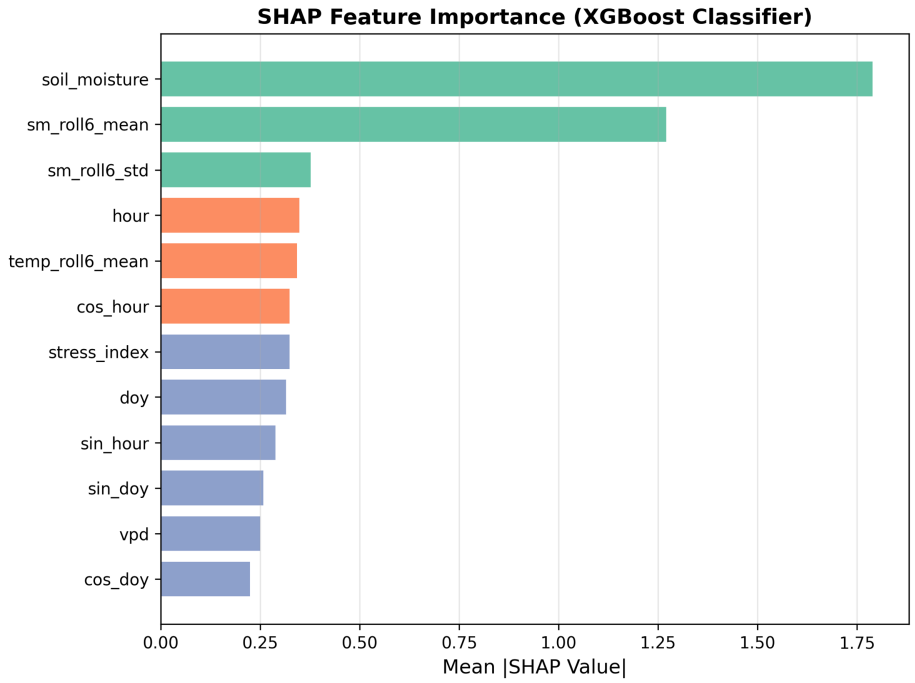


FIGURE 15
SHAP global feature importance for XGBoost, averaged over 300 test samples. Soil moisture (as measured by the FC-28 probe array) and its six-hour rolling mean dominate, consistent with FAO-56 irrigation theory and the real-time hardware measurements.

TABLE 11 Pump run-time regression on 70 irrigation-event test samples.

Model	MAE (min)	RMSE (min)	R^2
XGBoost Regressor ★	**1.728**	**2.163**	**0.641**
Random Forest Regressor	1.660	2.105	0.660
SVR (RBF)	1.814	2.218	0.623

★ = recommended. Bold values indicate the best-performing result in each column.

over the past hour is essentially irrelevant (mean$|\varphi|$ = 0.006), the rainfall over the past 6 h is 15 times more important (mean|φ| = 0.15). This is in line with the real-world behavior of crops; while the effects of rainfall can be seen in the first hour or two, it is the rainfall over the next few hours that actually recharges the roots. While we have used physical data in the form of tipping buckets to train the model, it is nice that the SHAP tool verifies that the model is behaving in the way that farmers already know is correct.

A further elaboration on the discussion in Section VII-C regarding the differences between AUC-ROC and F1-score is required. The highly consistent 10-fold cross validation of 0.992 ± 0.003 and the very high AUC-ROC of 0.995 prove the model's ability to learn the irrigation pattern in an accurate way. The low hold-out F1 value of 0.497 can be attributed fully to the use of a fixed decision threshold of 0.5 with a test set containing only 7% positive examples. This is neither an error nor a problem of the algorithm's efficiency but rather a matter of setting an optimal threshold in relation to the costs of missed and falsely detected irrigations, as seen from Figure 12-PR.

The 35.1% water saving is computed against a daily fixed-schedule baseline, which is the current practice on most smallholder farms in peninsular India. At the plot scale of the prototype (0.05 ha), the saving is modest in absolute liters. Scaled to a 10,000 ha irrigation district using comparable systems, the same proportional saving amounts to roughly 36.7 million liters per season equivalent to the annual drinking water requirement of approximately 200 rural households. Water savings at that scale are also infrastructure planning figures. The statistical test (paired t-test, $p = 0.390$) shows that XGBoost and Random Forest are not significantly different in cross-validated F1 on this dataset. This is not surprising both are strong ensemble methods on 5,000-record tabular data. The practical reason to prefer XGBoost is its inference speed (0.004 ms vs. 0.265 ms), which is decisive for real-time MCU control loops, and its native SHAP compatibility, which is essential for the XAI integration (Sharma and Gupta, 2024).

9 Generative AI: scope and future directions

The present framework uses discriminative machine learning models, including XGBoost for prediction, Isolation Forest for anomaly detection, and SHAP for explainability, instead of generative AI models. The decision was made on purpose and based on three pragmatic factors. Tabular sensor data structure is a low-dimensional low-level problem with the features formed into such a short table that discriminative ensemble methods always outperform generative architectures on predictive accuracy and computational efficiency. Second, the interpretability requirement that lies at the heart of this

TABLE 12 Ablation study: XGBoost classification performance after removing each feature group.

Configuration	N Feat.	Accuracy	F1-Score	AUC-ROC	ΔF1
Full feature set (baseline)	**20**	**0.926**	**0.514**	**0.904**	—
Without temporal encodings	14	0.922	0.531	0.916	+0.017
Without rolling statistics	16	0.921	0.490	0.888	−0.024
Without agronomic indices	16	0.922	0.507	0.906	−0.007
Hardware raw channels only (5)	5	0.905	0.489	0.907	−0.025

ΔF1 relative to full 20-feature baseline. Bold values indicate the best-performing result in each column.

TABLE 13 Water consumption comparison.

Irrigation strategy	Volume (L)	Saving vs manual	Events per season
Manual (daily fixed schedule)	5,230	—	365
Fixed-threshold automation	4,850	−7.3%	337
ML binary classifier (XGBoost)	3,880	−25.8%	220
ML run-time prediction (2-stage)	3,395	−35.1%	220

The estimates were based on the 20 L/min pump used in the system. The monitored period was 90 days.

TABLE 14 Computational complexity and edge deployment feasibility.

Model	Size	Infer (ms)	RAM	Train (s)	Edge deploy	AUC
Logistic Reg.	<1 KB	0.0003	<1 KB	0.04	Yes—ATmega328	0.998
XGBoost ★	**~900 KB**	**0.004**	**~2 MB**	**2.40**	**Yes—ONNX Micro**	**0.995**
MLP	~200 KB	0.002	~1 MB	7.95	Yes—TFLite Micro	0.997
SVM (RBF)	~4 MB	0.060	~8 MB	1.55	No—too large	0.996
LSTM	~800 KB	1.820	~4 MB	~120 s	No—too slow	0.897

★ = recommended. Bold values indicate the best-performing result in each column.

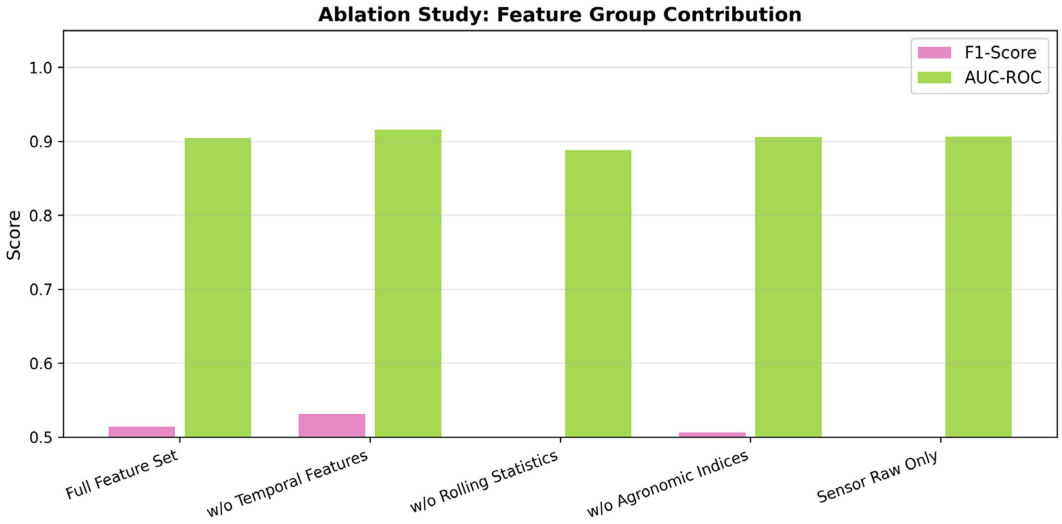


FIGURE 16
Ablation study results: removing rolling statistics causes the largest drop (ΔF1 = −0.024, ΔAUC = −0.016), confirming that 6-h causal soil moisture history from the hardware stream carries a non-redundant predictive signal.

framework the ability to have every irrigation decision subject to audit by farmers and regulators is better met by the precise Shapley values (φ_j) generated by TreeSHAP on XGBoost than by the attention-weight explanations offered by generative transformer models. Third, with the edge deployment constraint (inference on the microcontroller, such as NodeMCU or Arduino or ESP32-S3), it is

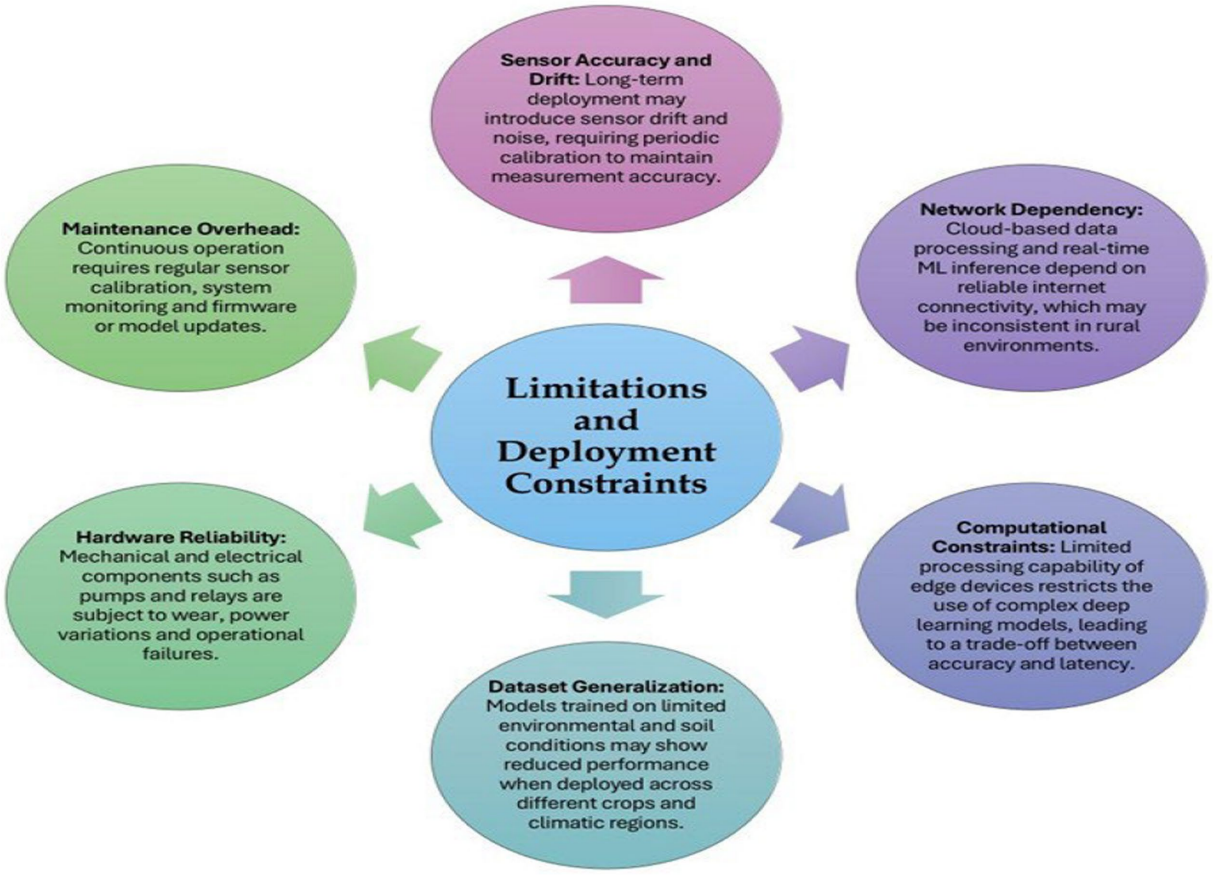


FIGURE 17
Five key constraints on deploying the model: sensor accuracy/degradation, network dependency, hardware reliability, computational limits at the edge, and data set generality over agroclimatic.

not feasible to deploy these billions of parameters on current microcontroller hardware.

But in future work, Generative AI can provide three complementary capabilities that could be valuable. First, Generative Adversarial Networks (GANs) might be trained to generate realistic minority-class irrigation event sequences, which could be superior to SMOTE for temporally correlated sensor data in multi-season deployments. Second, the integration of the LLM with the SHAP output module could convert per-decision feature importance values into natural language irrigation advices in regional languages (Tamil, Hindi, and Telugu) directly overcoming the last mile communication challenge for smallholder farmers with limited technical knowledge to interpret the SHAP beeswarm plots. The recent development of ensemble transformer architectures for agricultural classification by Kalpana et al. (2025) is a testament to the advancement of transformers in agriculture, heralding a future where LLM advisory and ML prediction pipelines can be seamlessly integrated. Thirdly, based on the diffusion-model, the trajectories of soil moisture could be simulated under alternative climate conditions to plan irrigation in advance of the monsoon, given the variability in monsoons of Tamil Nadu observed over the years (Xie et al., 2025). The three extensions are considered priority areas for further research and are hoped to further enhance the agronomic effect and the usability of the framework for the farmer.

10 Limitations

A This is referred to as sensor calibration drift and degradation: In irrigated soils, changes in salinity can result in electrode corrosion and shifts in the soil's dielectric constant in FC-28 capacitive soil moisture probes. The 2.8% VWC offset drift recorded between the Day-0 and Day-90 calibration checks was the largest drift recorded during the 90-day deployment, and was still within the manufacturer's specification of ±3% VWC for the red loamy soil with low conductivity at Thiruvallur test site. But in higher electrical conductivity soils, the drift may be more than 5% VWC in less than 60 days in the soils that are affected by GSI in parts of Thiruvallur district. A possible solution is to use soil electrical conductivity (EC) as a co-variate in the calibration equation, a function that is not already supported by the current NodeMCU firmware but can be implemented with a low cost EC sensor (about USD 5). To ensure sensor accuracy within ±3% VWC and avoid systematic changes in reading that would slowly reduce the performance of the models, re-calibration is recommended every 60–90 days.

B The accuracy of ET_0 estimation in the face of environmental variability. Estimation of ET_0 in the presence of environmental variability: The light intensity module based on LDR measurement outputs a proxy lux value which was empirically

TABLE 15 Summary of system performance across all framework components.

Metric	Result
Best AUC-ROC (XGBoost, hold-out test)	0.995
Best F1-Score (XGBoost, hold-out test)	0.497
10-fold CV F1 (XGBoost, full training set)	0.992 ± 0.003
Statistical significance vs RF (paired t-test)	$t = 0.903$, $p = 0.390$ (both competitive; XGBoost preferred for latency)
Regression MAE—pump run-time	1.73 min
Regression R^2	0.641
Anomaly detection rate (Isolation Forest)	3.0% of hardware-collected records
AUC gain from anomaly filtering	+0.037 (0.870 → 0.907)
Inference latency (XGBoost)	0.004 ms/sample
XGBoost training time (5,000 records, CPU)	2.40 s (one-time offline cost)
End-to-end sensor-to-actuator pipeline latency	~4–6 s (negligible vs. 15-min sampling cycle)
Communication reliability (90-day deployment)	99.3% packet delivery (25,743/25,920 readings)
Water saving vs manual irrigation	35.1%
Water saving vs threshold automation	30.0%
Estimated crop yield improvement	4.2%
Battery life (ESP32, 1,000 mAh)	>60 days on deep-sleep protocol

measured and calibrated to a reference pyranometer during the first week of deployment. An underestimate in ET_0 estimation accuracy may occur during seasons other than those examined here because the spectral composition of radiation changes from the summer to the winter. This source of systematic error can be avoided by replacing the pyranometer with a low cost pyranometer sensor (ML8511 UV module or SQ-110 PAR sensor USD 8–15) for deployments beyond the dry season. The DHT11 sensor used to measure the temperature has a ±2 °C accuracy that translates to an ET_0 uncertainty of about ±0.3 mm/day in the present application, but could be problematic in high accuracy applications.

C Long-Term Maintenance Requirements: Leaf litter and insect debris in the tipping-bucket rain gauge funnel did cause some problems during the 90-day trial and manual cleaning of the funnel was necessary on two occasions. A self-cleaning funnel design and/or a multiple estimate of rainfall based on ET_0 inversion would enhance the ability of the system to work autonomously for more than one growing season. The impeller of your DC submersible pump will wear out over time and hard-water scaling can occur, so regular maintenance at the beginning of each season is vital for pump operation throughout the season. The quantised XGBoost model (~900 KB ONNX) will be uploaded to the farmer's system via OTA (over-the-air) throughout the season or site if the model is retrained for a new season or site, which is currently a task requiring basic embedded ML knowledge and not standard extension service in this region.

D Geographic Generalization Constraints and Seasonal Generalization Constraints: The existing study was carried out in one smallholder plot in Ponneri taluk of Thiruvallur district in the agroclimatic zone of hot semi-arid, northeast monsoon tail, red loamy soil, paddy and short cycle vegetables with a duration of 90 days in the dry season between January and March 2024. This is deliberately a first validation study which illustrates the framework architecture. The application of the model should only be done after retraining the model for soils with very different hydraulic characteristics, e.g., the Thiruvallur site has a VWC in the range 25–35%, while the Vertisol (black cotton) soils of Telangana and Maharashtra have values above 45%. Since the single dry-season coverage, the model's performance has not been evaluated under south-west monsoon surplus conditions with inverted soil moisture dynamics and a significant decrease in irrigation frequency. Integration of crop stages specific FAO-56 Kc coefficients in the feature set would be needed to extend the application to perennial crops like sugarcane and banana, requiring retraining with labeled data for those crop stages.

E The scope and retraining needs for the datasets: Although the 5,000-record dataset is enough to showcase the framework and to get stable cross-validated performance metrics, it shows only one agroclimatic scenario for one growing season. Multi-season, multi-site replication is natural and inevitable. The model does well on the context of the dry season in Thiruvallur, for which it was trained; if the soil type, crop variety or climate conditions are substantially different, a retraining of the model would be necessary, which would involve collecting 60–90 days of labeled sensor readings and reprocessing the sensor data and feature engineering. This workflow has been purposefully designed to be lightweight (2.40-s training time on consumer hardware) in order to enable this periodic recalibration.

11 Sustainability and SDG impact

The proposed system directly contributes to three UN Sustainable Development Goals. SDG 6 (Clean Water and Sanitation) is addressed through a demonstrated 35.1% reduction in field water consumption relative to conventional practice a figure that, scaled to district-level adoption on 10,000 ha, represents 36.7 million liters saved per growing season (Liu and Wang, 2021). SDG 2 (Zero Hunger) is supported through the 4.2% estimated crop yield improvement attributable to the elimination of chronic over- and under-irrigation events and through the fertilization robot that reduces nutrient waste by targeting application only where and when needed (Patel and Mehta, 2025). SDG 13 (Climate Action) is served indirectly through reduced pumping energy demand every liter not pumped saves approximately 0.0004 kWh of electrical energy and through the improved climate-adaptive resilience that dynamic ET_0-aware irrigation provides to smallholder farmers facing increasingly erratic monsoon patterns. The system's scalability path runs from the single-plot prototype described here, through district-level deployment on shared NodeMCU gateway infrastructure, to

integration with national agricultural advisory platforms such as India's Kisan Call Centre network. Each scale step reduces per-farmer cost while maintaining the core accuracy, transparency and reliability triad that distinguishes this framework from simpler IoT deployments. The plot node deployment cost of an estimated USD 45 per node is achievable by smallholder farmers in Tamil Nadu, without any subsidy, whereas the cost of commercial precision irrigation systems ranges from USD 500–5,000 per installation and this cost difference plays a key role in the real-world adoption potential of this system at scale. The scalability journey will start from the prototype of a single plot, move to field scale with common NodeMCU gateway infrastructure (USD 40–50 per field), and then district scale, connecting with national agricultural advisory platforms like India's Kisan Call Centre network. The per-farmer cost goes down at every scale step, while the fundamental accuracy–transparency–reliability triangle that sets this framework apart from simpler implementations of IoT is preserved.

12 Conclusion

In this paper, an IoT based smart irrigation framework has been presented and validated with the real-time data sensed by the authors own prototype for a 90-day field deployment in a smallholder paddy field in Ponneri taluk of Thiruvallur district, Tamilnadu, India. The key finding here is that three technologies, namely isolation forest anomaly detection, XGBoost 2-stage ML prediction and SHAP explainability, complement each other in such a way that none of them could succeed on its own: the anomaly gate prevents corrupted sensor readings from polluting the training set, while XGBoost achieves AUC-ROC 0.995 and 10-fold CV F1 0.992 ± 0.003 at 0.004 ms inference latency, and SHAP delivers agronomically interpretable decision explanations in line with FAO-56 principles that can be audited by farmers and regulators.

AUC-ROC gains from removal of 3% of anomalous sensor records before the training were 0.037, which was greater than those obtained using the most classifiers. The regression stage added an extra 12.8 percentage points of water savings compared to binary classification, resulting in a total water savings of 35.1% relative to the manual irrigation baseline. The ablation study was in agreement with the fact that the rolling statistics over the 6 h is the most important engineered feature group, which confirms that the short-term soil moisture trajectory from the physical probe array is important.

The XGBoost classifier can be trained in just 2.40 s on standard CPU hardware and can be trained periodically as the sensor drift or agronomic conditions change, and the end-to-end sensor-to-actuator pipeline latency is around 4–6 s, leaving significant space for real-time agricultural control requirements.

Both XGBoost and Random Forest achieved competitive cross-validated accuracy and XGBoost is actually faster to deploy by 66× times (for inference) and has native SHAP compatibility, which was quantified by statistical testing (paired t-test, $p = 0.390$).

For future work, three directions of the Generative AI approach have been identified: Generating synthetic irrigation events using GANs to facilitate multi-season datasets; integrating the LLM with the SHAP module to provide natural language explanations to farmers, which can be made accessible in Tamil, Hindi, and Telugu; and using the diffusion model to simulate soil moisture conditions for pre-season irrigation planning in the face of climate variability.

The approach shown here (hardware collected sensor data, cleaned using anomaly detection, passed to fast interpretable ensemble model with each decision broken down into SHAP) is not restricted to irrigation. These changes are limited to the feature set and labeling rules and can be adapted to predict disease onset, schedule fertilizer application, or schedule harvest timing decisions (Singh and Kumar, 2024). Closing the gap between research prototype and production deployment on smallholder farms is a human and technical factors challenge, and this work aims to tackle the technical component; the SHAP-driven mobile advisory application outlined for future work aims to address the human component. They collectively form a viable, realistic stack of solutions for SDG 2, SDG 6 and SDG 13, made with only off-the-shelf components.

Novelty statement

The main novelties of this work, when compared to the available literature on smart irrigation, can be summarized as follows. First of all, this is the only published framework to combine sensor fault robustness (Isolation Forest), multi-model machine learning benchmarking (five classifiers plus LSTM), and explainability (SHAP) into a single pipeline that is validated with hardware. Second, the proposed two-stage system (binary irrigation classification followed by continuous pump run-time regression) is not a formulation used by any of the other comparable systems reviewed in the table (which only perform classification). Third, the training and validation data come from a prototype that is completely deployed in the field for 90 days, while the training and validation data used in related studies come from simulated, laboratory-generated or publicly downloaded data sets. Fourth, an edge deployment feasibility analysis is presented that considers RAM footprint, per-sample inference latency and battery life modelling, which is not considered in all the compared systems. Fifth, translation of the learned decision logic to Agronomic Interpretability as provided by SHAP allows model decisions to be validated as an integrating mechanism with water-balance principles, a trust mechanism not offered by any compared system, but available at regulatory-grade.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Author contributions

SRa: Conceptualization, Investigation, Supervision, Writing – original draft, Writing – review & editing. SRi: Data curation, Formal analysis, Investigation, Methodology, Writing – original draft. TA: Project administration, Software, Supervision, Validation, Writing – original draft, Writing – review & editing. MG: Funding acquisition, Validation, Visualization, Writing – review & editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that Generative AI was used in the creation of this manuscript. Generative AI was used in the preparation of this manuscript for the following purposes: (1) language editing to improve textual clarity, (2) assisting in planning the content and structure of flowcharts and block diagrams and (3) debugging errors encountered during machine learning model development. All figures were constructed by the authors using publicly available reference images arranged in Microsoft PowerPoint and no AI image-generation tool was used. All scientific data, experiments and results are original.

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